16/02/26

32 - Hydroxy compounds

Alcohols

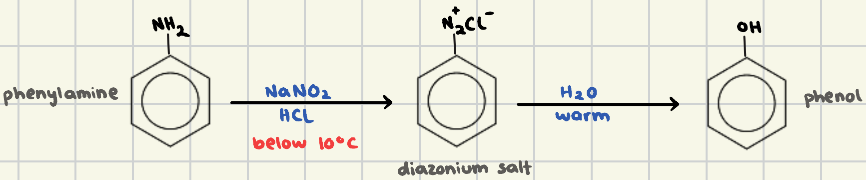
alcohol + acyl chloride ⟶ ester + hydrogen chloride $_{(g)}$

e.g. $C_2H_5OH + CH_3COCl \longrightarrow CH_3COOC_2H_5 + HCl$

ethanol, ethanoyl chloride, ethyl ethanoate

Phenols

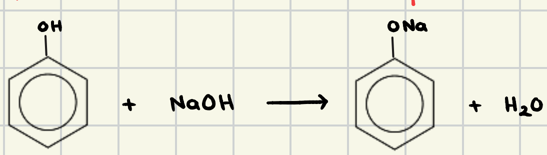
Production of phenol :



Reactions of phenol : (the reaction of other phenolic compounds are similar to those of phenol)

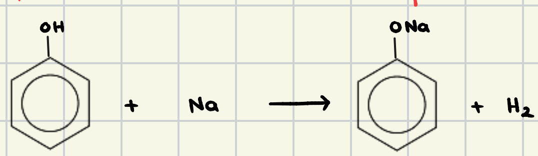
1) reaction with bases

phenol + base ⟶ sodium phenoxide



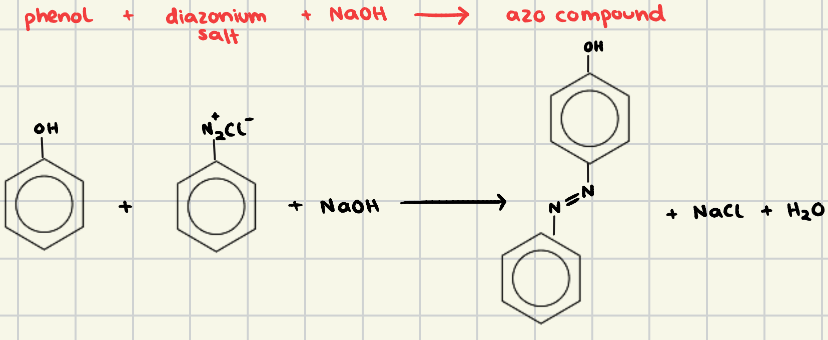
2) reaction with Na $_{(s)}$

phenol + sodium ⟶ sodium phenoxide



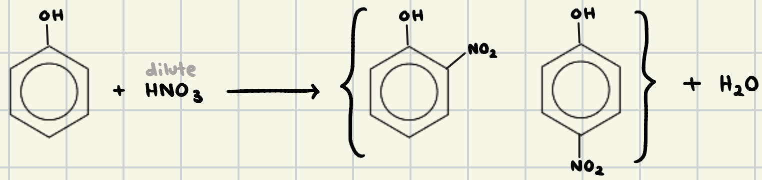
3) reaction with diazonium salts in NaOH

phenol + diazonium salt + NaOH ⟶ azo compound



4) nitration of the aromatic ring

phenol + dilute HNO_3 ⟶ 2-nitrophenol + (mixture of) 4-nitrophenol



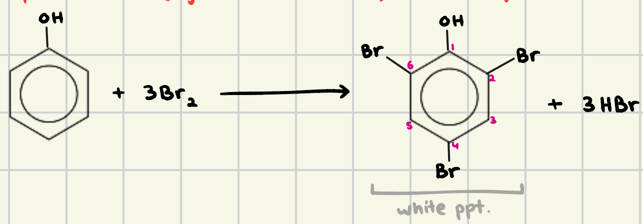
(this reaction does NOT need a catalyst unlike benzene)

phenol undergoes nitration & bromination MORE readily than benzene, due to the activating effect of the hydroxyl group, which increases the electron density in the ring, making it more reactive to electrophiles

benzene requires stronger conditions for these reactions while phenol reacts under milder conditions

5) bromination of the aromatic ring

phenol + $Br_{2\,(g)}$ ⟶ 2,4,6-tribromophenol

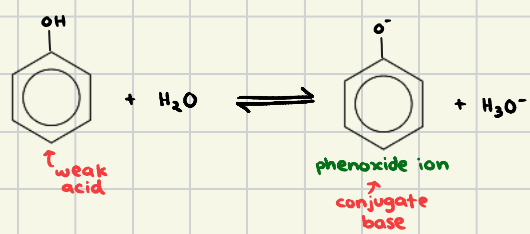


(this reaction does NOT need a catalyst unlike benzene)

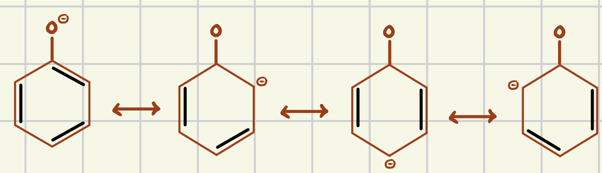
the hydroxyl group (-OH) in phenol is an e^--donating group & directs substitution to the 2-, 4-, 6-positions on the aromatic ring due to resonance & e^- density effects

Acidity of Phenol

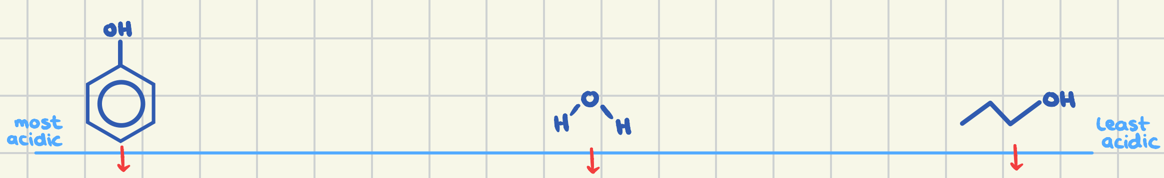
→ phenol is a weak acid because it can lose a H^+ from the hydroxy group



* The phenoxide ion is more stable than phenol
* The ion is stabilised by resonance
* As the -ve charge is delocalised over the aromatic ring
* Making phenol more acidic than alcohols but less than carboxylic acids



Acidity of phenol VS water VS ethanol



most acidic because phenoxide ion is stabilised by resonance with the aromatic ring

more acidic than ethanol but less than phenol, as it does not benefit from resonance stabilisation

least acidic as the ethyl group is e^- donating and does not stabilise the alkoxide ion ($C_2H_5O^-$)